

Tanishka Ganesh Mali

State College, PA | +1 (814) 280-6430 | tanishkamali114@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

Penn State University, University Park	Aug 2024–May 2026
Master of Science: Cybersecurity Analytics & Operations	GPA: 3.93
Pune University	Aug 2021–May 2024
Bachelor of Engineering: Computer Engineering	GPA: 3.56

Work Experience

Penn State University–Amazon	Jan 2026–May 2026
<i>Agentic AI Developer, State College PA</i>	

- Built an agentic AI red-teaming framework using Strands SDK, AWS Bedrock, and Python to orchestrate multi-agent workflows for automated web application security testing, vulnerability discovery, and validation across development environments.
- Developed Playwright and TypeScript-based automated testing suites to simulate attacker and benign-user workflows, improving vulnerability validation, regression detection, and feature correctness by **40%+**.

Balbird Industries	Jun 2023–Dec 2023
<i>Vulnerability Management Intern, Pune IN</i>	

- Performed vulnerability assessments on ERP systems, validating OpenVAS findings through XSS and input validation exploits, reducing risk by **40%**.
- Ensured compliance with **IT Act (2000)** and **PDPB** by enforcing secure password and lockout controls; applied data minimization to reduce log exposure by **60%**, documenting remediation evidence & escalation status for management review.

Keenhour (99 Digital)	Feb 2023–May 2023
<i>Software Developer Intern, Pune IN</i>	

- Improved application performance by **40%** through code splitting and lazy loading, optimizing frontend-backend interactions, reducing initial bundle size, and supporting batch processing and asynchronous workflows.
- Enhanced API reliability by implementing robust HTTP request handling (Axios with retry/backoff), reducing failure rates by **30%** and improving end-to-end system stability.

Projects

Container Security Pipeline with Binary Authorization (GCP) [Github]	Aug 2025–Nov 2025
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- Designed and implemented a secure container CI/CD pipeline using GCP Binary Authorization, KMS asymmetric signing, and Grafeas Attestors; enforced supply-chain security controls and reduced deployment of unsigned container images by **100%**.
- Automated Kubernetes image validation by enforcing digest verification and attestation policies to DENY unsigned images and ALLOW KMS-signed artifacts in GKE workflows, improving deployment integrity and policy compliance.

AI-Based Network Intrusion Detection System [Github]	Jan 2026–May 2026
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- Built an AI-based NIDS on CIC-IDS2017 using supervised models (Random Forest, Gradient Boosting) and unsupervised models (Isolation Forest, LOF, OCSVM), achieving **98.8% precision**, **>0.99 ROC-AUC**, and stable detection on **100K+ network flows**.
- Engineered a Streamlit dashboard with real-time metric slicing, automated PR/ROC visualization, and a hybrid anomaly engine combining ML scores with signature-style rules, reducing manual triage effort by **40%**.

IoT Threat Mapping using MITRE ATT&CK [Github]	Jan 2024–May 2024
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- Developed an IoT threat mapping framework aligning IoT/ICS attack scenarios to MITRE ATT&CK techniques, enabling structured threat modeling and improving detection coverage across **15+ adversary behaviors**.
- Engineered detection mappings and Sigma-style rules for IoT telemetry, supporting threat-informed defense and reducing analysis gaps across mapped TTPs.

Certifications

CompTIA Security+ (SY0-701)	Sept 2025
AWS Certified Solutions Architect – Associate	In Progress

Skills

Security: Threat Detection, Threat Hunting, Incident Response, DFIR, Vulnerability Assessment, Application Security, Secure SDLC, Threat Modeling, Detection Engineering, MITRE ATT&CK, OWASP Top 10, IAM, System Hardening, NIST SP 800-53, SSL/TLS
Cloud & Infrastructure: AWS (S3, Bedrock, IAM), GCP (Binary Authorization, KMS, GKE), Docker, Kubernetes, CI/CD Security, Container Security, Supply Chain Security, Grafeas Attestors, Prometheus, OpenTelemetry
Tools & Platforms: Splunk, Wazuh, Burp Suite, OpenVAS, Nessus, Wireshark, Ghidra, IDA Pro, Git, Jenkins, Playwright
Programming & Automation: Python, Bash, JavaScript, TypeScript, C/C++, SQL
AI Security: LLMs, Amazon Bedrock, Agentic AI, Adversarial AI, Prompt Engineering, Model Evaluation, Anomaly Detection

Competitions

INJ Cyber Competition – Incident Response Simulation	2nd Place, 2025
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- Detected CVE-2021-4034 exploitation and credential abuse, reconstructed attack timelines, and contained attacks across SSH, RDP, MySQL, and LDAP services.

Raymond James Intercollegiate CTF <i>[Best Team Spirit Award]</i>	8th Place, 2025
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- Identified S3 misconfiguration and RAG data-poisoning risks; automated challenge solving across **1K+ inputs**.

DASSH Homeland Security Design Challenge [Github]	2025
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- Built Python/Wazuh-based AI log analysis for **10K+ security events**, reducing false positives by **60–70%**.